BANKING INFORMATION SYSTEM

Core Java Internship & Project Report

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Project: Banking Information System

Technology: Core Java

Repository: upskillcampus

USC / UCT Report Format

1. Introduction

The Banking Information System is a console-based Java application developed as part of the Core Java internship project. It demonstrates basic banking operations through a simple menu-driven interface.

2. Objectives

• Create a basic banking application using Java.
• Practice classes and objects.
• Apply encapsulation using private data members.
• Implement deposit, withdrawal and balance-checking operations.
• Practice user input and menu-driven programming.

3. Technologies Used

Programming Language: Java
Concepts: Classes, Objects, Encapsulation, Constructor, Methods, Scanner, Conditional Statements and Loops.
Execution: JDK / Java compiler and terminal.

4. Main Features

1. Display Account Details
2. Deposit Money
3. Withdraw Money
4. Check Balance
5. Exit

5. Project Working

The program first accepts the account number, account holder name and initial balance. It then displays a menu. The user selects an operation, and the corresponding method performs the requested action. The menu continues until the user selects Exit.

6. Core Java Concepts Applied

Class and Object: A BankAccount class stores account information and an object represents the account.

Encapsulation: Account data members are private and accessed through methods.

Constructor: The constructor initializes account information.

Methods: Separate methods handle deposit, withdrawal, balance checking and account display.

Scanner: Used to receive input from the user.

7. Testing and Output

The program was executed in the terminal. During testing, account details were entered and the menu was displayed. An invalid menu input was also tested; when 500 was entered where a menu choice was expected, the program displayed “Invalid choice.”

8. Sample Output

```
==============================
BANKING INFORMATION SYSTEM
==============================
Enter Account Number: 100
Enter Account Holder Name: LAKSHAY
Enter Initial Balance: 500000

========== MENU ==========
1. Display Account Details
2. Deposit Money
3. Withdraw Money
4. Check Balance
5. Exit
==========================
Enter your choice: 500
Invalid choice.
```

9. Challenges and Solutions

One challenge was understanding how the menu choice should be entered separately from the transaction amount. This was handled by following the menu step first and then entering the amount only after selecting Deposit or Withdraw.

10. Learning Outcomes

This project improved practical understanding of Java syntax, object-oriented programming, constructors, methods, encapsulation, Scanner input, loops and conditional logic. It also provided practice in compiling and running a Java program from the terminal.

11. Conclusion

The Banking Information System demonstrates a simple banking workflow using Core Java and provides practical implementation of basic object-oriented programming concepts and menu-driven application logic.

12. GitHub Submission

Repository name: **upskillcampus**
Code file: **BankingInformationSystem.java**
Report file: **BankingInformationSystem_Lakshay_USC_UCT.pdf**

The actual GitHub URLs should be inserted after the repository is created.